Object embedding vector generation

Object embedding vector generation

1. NVIDIA official application
2. Case goal and processing chain
   2.1 People feature vector
   2.2 Retail product feature vector
3. Startup entry and output
4. Local video recognition
   4.1 Run
   4.2 Replace the local video
   4.3 Why the retail video preview plays slowly frame by frame while `out.mp4` is smooth
5. Real-time recognition with the USB camera
   5.1 Run
   5.2 About Retail
6. Real-time recognition with the CSI camera
   6.1 Run
7. Stopping, video output and troubleshooting

> Applicable environment: JetPack 7.2.1, Ubuntu 24.04, DeepStream 9.1, Jetson Orin NX 16GB.

> The factory environment already has the MDX application, the models, the TensorRT engines and the course entry. Start from section 4 for daily learning; refer to section 1 to deploy again only when the models, the engines or the program are damaged.

1. NVIDIA official application

This case uses the official NVIDIA DeepStream `deepstream-mdx-perception-app`, whose source code is located at:

text

```
~/deepstream_tao_apps/apps/tao_others/deepstream-mdx-perception-app
```

Official documentation: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps/apps/tao_others/deepstream-mdx-perception-app

The focus of MDX is not only the detection boxes, but the **embedding (feature vector)** attached to the targets. A feature vector is a sequence of numbers that can be used for later matching of same-class targets, retrieval of similar targets, cross-view association or database analysis.

The current course environment has already completed the application build, the model download and the engine build. Do not delete the `.engine` files under `models/` casually; the engines are tied to the current Jetson, CUDA and TensorRT environment, and after the deletion the first run rebuilds them and takes a long time.

2. Case goal and processing chain

The course provides two MDX chains: one for people and one for retail products.

2.1 People feature vector

```
PeopleNet Transformer person detection
        ↓
NvDCF Tracker object tracking
        ↓
ReIdentificationNet
        ↓
Person embedding (256 dimensions)
```

The people mode is usually faster than the retail mode. The reason is that the people detection model is lighter, and the people targets in the frame are generally fewer than the product targets on the shelves.

2.2 Retail product feature vector

text

```
Retail DINO product detection
        ↓
NvDCF Tracker object tracking
        ↓
Retail Object Recognition
        ↓
Product category + embedding (448 dimensions)
```

The retail mode performs a second-level recognition on every detected product and extracts a feature vector. The input of Retail DINO is `960×540`, which is clearly heavier than the people detection, so real-time camera scenarios need a dedicated low-latency strategy.

3. Startup entry and output

The current unified startup entry:

```
~/deepstream_tao_demo/run_mdx.sh video
```

Input source	People mode	Retail mode
Local video	`~/deepstream_tao_demo/run_mdx.sh video`	The Retail Object Recognition model is needed (it is not deployed in this environment)
USB camera	`~/deepstream_tao_demo/run_mdx.sh usb`	The same, with the retail YAML
CSI camera	`~/deepstream_tao_demo/run_mdx.sh csi`	The same, with the retail YAML

All three entries display a window on the desktop of the Jetson device itself or on the VNC desktop. Parameter description:

- `-c`: the YAML configuration file
- `-m 2`: the ReID pipeline (`-m 3` is the Retail one)
- `--tiledtext`: display the tiled text

4. Local video recognition

4.1 Run

```
cd ~/deepstream_tao_demo
export DISPLAY=:0
./run_mdx.sh video
```

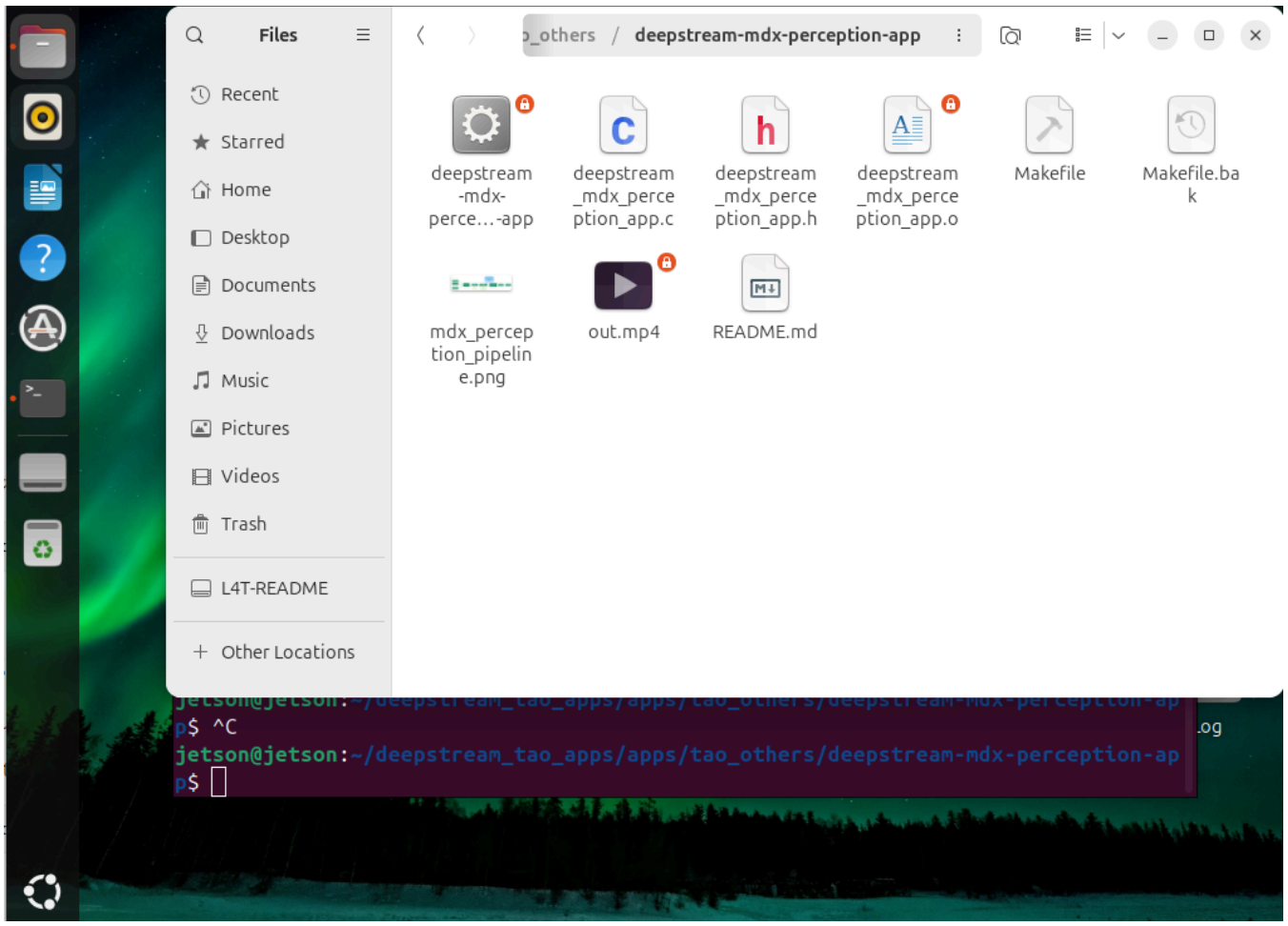


The video entry displays the detection results in real time and generates an `out.mp4` with the detection boxes in the MDX application directory:

```
~/deepstream_tao_apps/apps/tao_others/deepstream-mdx-perception-app/out.mp4
```

After the run you can directly check whether the file is complete:

```
ffprobe -v error -show_entries format=duration -of default=nw=1 \
  ~/deepstream_tao_apps/apps/tao_others/deepstream-mdx-perception-app/out.mp4
```



4.2 Replace the local video

The people mode modification:

```
~/deepstream_tao_apps/configs/app/sources.csv
```

Change the video URI in the CSV to the absolute path on the Jetson:

```
1,3,file:///home/jetson/Videos/test.mp4,1,0,0
```

Using MP4/H.264 files is recommended. The Jetson path must be entered here; a Windows path must not be entered.

4.3 Why the retail video preview plays slowly frame by frame while `out.mp4` is smooth

The speed of the complete chain of the retail model is lower than the frame rate of the original video. For example, when the source video is 30 FPS and Retail DINO plus the second-level recognition can only process it slowly, the file source can be **back-pressured** by the downstream pipeline: the program slows down the file reading and advancing speed while it waits for the previous frame to finish the inference.

```
Local video file → inference slows down → file reading slows down accordingly → each frame is processed in turn
```

Therefore the preview window looks like a slow frame-by-frame playback, but it is still processing the complete video frames in order. The output MP4 keeps the original video timestamps and the player plays it on the normal timeline, so `out.mp4` looks continuous and smooth.

This is not “automatically reducing the number of pushed-in frames” or actively dropping video frames; the file input can wait for the inference to complete. For a local video the cost is a longer total processing time; the same method cannot be used for a real-time camera, because the camera does not pause its output.

When you stop early during a local video, press `Ctrl+C` once and wait for the EOS finishing message to appear in the terminal before you exit. The program lets the MP4 muxer finish writing the `moov` index; do not press `Ctrl+C` repeatedly, close the terminal directly or use `kill -9`, otherwise `out.mp4` may not be playable.

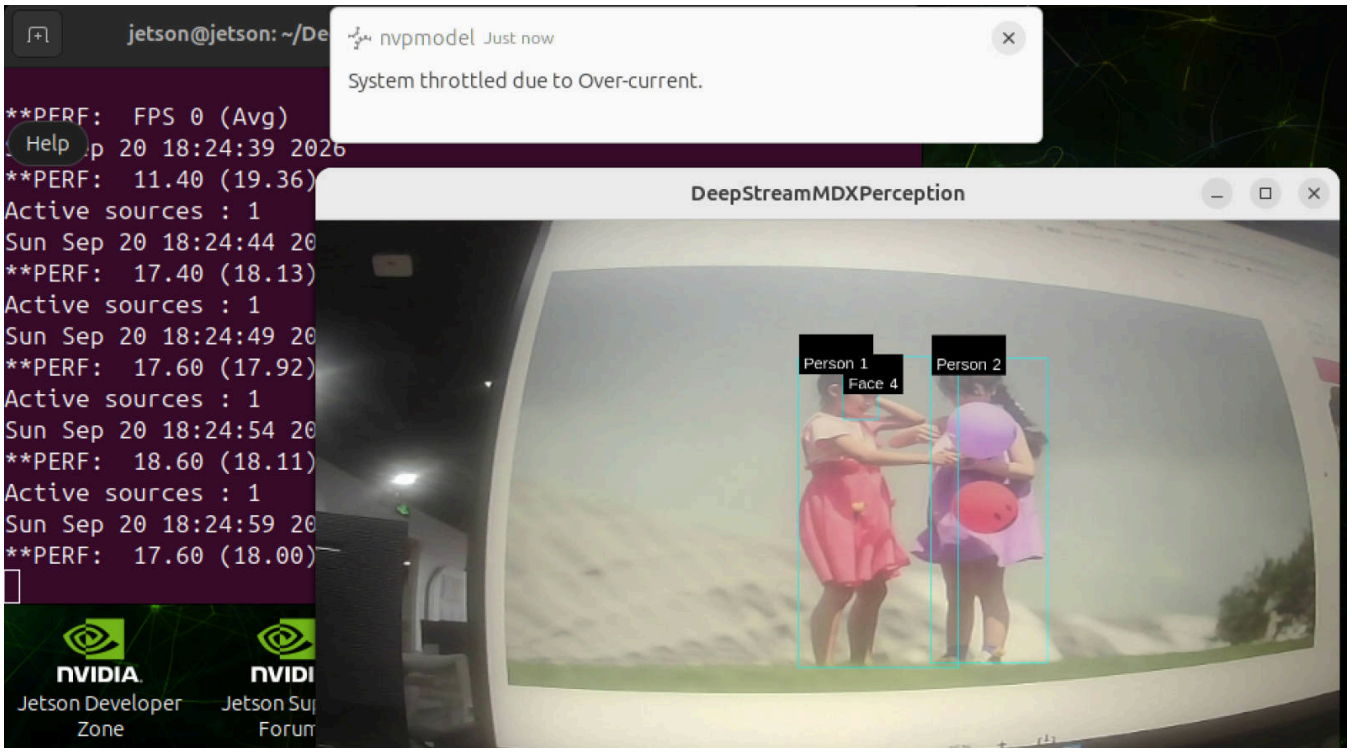
5. Real-time recognition with the USB camera

5.1 Run

```
cd ~/deepstream_tao_apps/apps/tao_others/deepstream-mdx-perception-app
export DISPLAY=:0

sudo -E ./deepstream-mdx-perception-app \
  -c ../../../configs/app/peoplenet_reidentification_usb.yml \
  -m 2 --tiledtext
```

The current USB mode uses `sources_usb.csv` by default:




```
enable,type,camera-width,camera-height,camera-fps-n,camera-fps-d,camera-v4l2-dev-node
1,1,640,480,15,1,0
```

That is `/dev/video0`, 640×480, 15 FPS.

5.2 About Retail

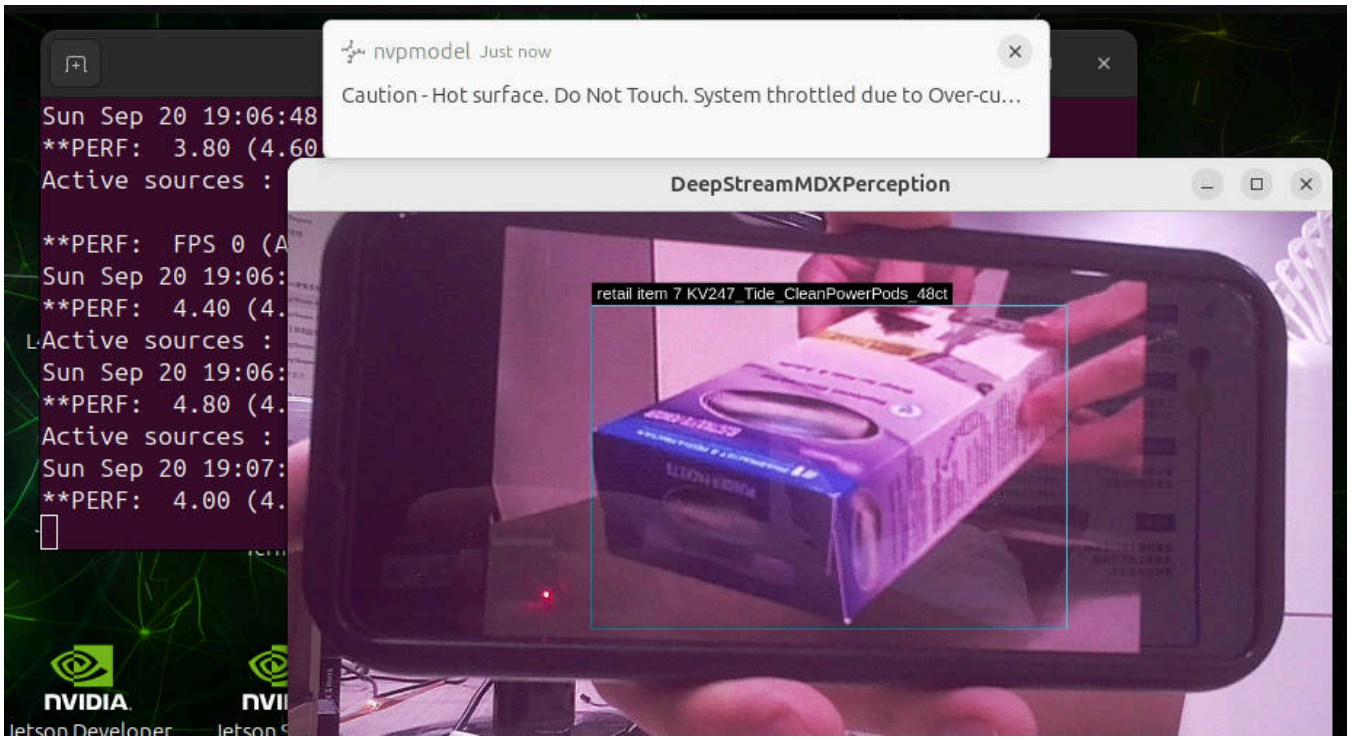
The current course entry does not have the Retail Object Recognition model deployed yet, so `-m 3` cannot be used directly. If you need to test the Retail pipeline, you must download the Retail Recognition model separately and put it under `~/deepstream_tao_apps/models/`.

6. Real-time recognition with the CSI camera

6.1 Run

Run:

```
~/deepstream_tao_demo/run_mdx.sh csi
```



The script generates a temporary CSI CSV under `/tmp/deepstream_tao/` and does not modify the repository template:

```
enable,type,camera-width,camera-height,camera-fps-n,camera-fps-d,camera-csi-sensor-id
1,5,1920,1080,30,1,0
```

That is `sensor-id=0`, 1920×1080, 30 FPS. MDX directly uses the Argus `type=5` and does not wrongly read `/dev/video10`.

7. Stopping, video output and troubleshooting

The real-time USB/CSI mode by default only previews and writes JSONL, and does not generate an `out.mp4`. The local video mode generates an `out.mp4`.

When you stop a real-time camera or a file task:

1. Go back to the terminal where the startup command runs.

2. Press `Ctrl+C` only once.
3. Wait for `Received EOS. Exiting ...` or `App run successful` before closing the terminal.

Symptom	What to check first
No JSONL records in the people mode	Whether any people actually appear in the frame; no embedding is generated when there is no detected target.
No picture with USB	Run `v4l2-ctl -d /dev/video0 --list-formats-ext` and check the device number, the output format and whether it is occupied.
No picture with CSI	First confirm that the basic CSI case works, then check `sensor-id=0` and `nvargus-daemon`.
`out.mp4` cannot be played	Use `run_mdx.sh video`; the script cleans up the old file first and automatically runs `ffprobe` to verify the container after it exits.
`VNC shows nothing`	Use `sudo -E` to keep the DISPLAY environment variable; do not use a bare `sudo`.
It reports that the Retail Recognition model is missing	`-m 3` is not available; the current environment only runs `-m 2` (People ReID).